



11 Do not relax mid-shaft or ends during the retouch service, as this will cause overprocessing. Option: Oil may be applied to previously relaxed ends to protect from overprocessing caused by overlapping.

12 Rinse thoroughly with warm water to remove all traces of the relaxer. If the relaxer comes with a normalizing lotion or conditioner, comb it through the hair. Leave it on as indicated by the manufacturer and rinse thoroughly. Always follow the manufacturer's directions and the procedures approved by your instructor.

13 Shampoo at least three times with an acid-balanced neutralizing shampoo. If you are using an acid-balanced neutralizing shampoo with a color indicator, usually the color change will go from pink to white indicating all traces of the relaxer have been removed and the natural pH of the hair and scalp has been restored. Apply conditioner as per manufacturer's recommendations. Rinse.



14 Style the hair as desired.

CLEAN-UP AND DISINFECTION

- ☐ Clean and disinfect tools and implements.
- ☐ Clean and disinfect work area.
- ☐ Dispose of single-use items. Place all used linens, towels, and capes in the laundry.
- ☐ Wash your hands.

CURL REFORMING (SOFT-CURL PERM)



MATERIALS, IMPLEMENTS, AND EQUIPMENT

You will need all of the following implements, materials, and supplies:

- ☐ Applicator bottle
- ☐ Applicator brush
- ☐ Conditioner
- ☐ Disposable gloves
- ☐ End wraps
- ☐ Gentle clarifying shampoo
- ☐ Large-tooth comb
- ☐ Perm rods
- ☐ Plastic or glass bowl
- ☐ Plastic processing cap
- ☐ Protective base cream
- ☐ Tail comb
- ☐ Thio cream relaxer
- ☐ Thio neutralizing solution
- ☐ Thio waving lotion
- ☐ Waterproof cape



CAUTION

Hair that has been treated with hydroxide relaxers cannot be treated with thio products. The chemicals are not compatible!

PREPARATION

- 1 Wash your hands.
- 2 Clean and disinfect all tools and implements.
- 3 Conduct client consultation.

PROCEDURE

- 1 Perform an analysis of the hair and scalp. Perform tests for porosity and elasticity. Remember, this procedure requires that the hair and scalp be completely dry after completing the pre-service procedure. Most of the manufacturer's directions indicate a shampoo may be necessary before a soft-curl service; if so, drape the client for a shampoo and gently shampoo with a mild shampoo and towel-dry hair. Avoid irritating the client's scalp.
- 2 Re-drape the client for a chemical service.
- 3 Based on the manufacturer's recommendation for a preliminary strand test, conduct a strand test to determine proper timing and curl pattern prior to full-head application. Make note of the timing for the thio cream relaxer, strength used, and rod size.



4 Divide the hair into four sections. Clip the sections up to keep them out of the way and for better application of product. Apply a protective base cream to the hairline and ears. It is best to make $\frac{1}{4}$ -inch to $\frac{1}{2}$ -inch horizontal partings and apply protective base cream to the entire scalp.



5 Wearing gloves on both hands, begin application of thio cream relaxer to the most resistant area, usually at the back of the head and nape area. Using an applicator brush or tail comb, apply cream $\frac{1}{4}$ inch away from the scalp and topside and underside of the strand. Do not apply the cream to the ends of hair during this step. To avoid possible scalp irritation, do not allow cream to touch the scalp until the last few minutes of processing.



6 Repeat application in remaining sections. Apply this cream to the hairline last, since hair is most fragile in this region.



7 Review application of all four quadrants. If necessary apply more cream until all hair strands are covered. Apply cream to hairline and to ends of hair during this step.



8 After the thio cream has been applied to all sections, using an applicator brush, the back of the comb, or your hands, begin to smooth each section, starting at the first section where thio cream was applied. Never comb the cream through the hair.

9 Process according to the manufacturer's directions and strand test results for timing. Perform periodic strand tests until time has elapsed.

10 During the final remaining minutes of processing, apply the thio cream down to the scalp and through the ends of the hair, using additional cream as needed. Carefully smooth all sections using an applicator brush, your fingers, or the back of the comb.



11 Rinse thoroughly with warm water to remove all traces of the thio cream.



12 After rinsing the hair, towel-blot and part it into nine panels. Use the length of the rod to measure the width of the panels.



13A Wearing disposable gloves, apply thio wrap lotion to each section, and roll hair on the appropriate-sized perm rods. Begin wrapping at the most resistant area, usually the nape.



13B Make a horizontal parting the same size as the rod. Hold the hair at a 90-degree angle to the head. Using two end papers, roll the hair down to the scalp.

13B Position the rod half off-base. Option: Insert roller picks to stabilize the rods and eliminate any tension caused by the band.



13C Continue wrapping the remaining eight panels in numerical order using the same technique.



CAUTION

Maintain even saturation of the thio lotion as you work.



14 Place cotton strips around the hairline and neck to protect the client.

15 If the manufacturer or your instructor suggests using a plastic processing cap, cover all the hair completely. Do not allow the plastic cap to touch the client's skin.

16 Process according to the manufacturer's directions. Processing time will vary according to the strength of the product, hair type and condition, desired results, and strand test results. Check for proper curl development in 5-minute intervals.



17 When processing is complete, rinse the hair thoroughly for at least 3 minutes. Gently towel-blot each rod to remove excess moisture. Do not rub.

18 Re-drape client with fresh strip cotton around hairline and neck.



19 Measure approximately 6 to 8 ounces of the thio neutralizer in an applicator bottle, using more or less as needed. Slowly and carefully apply to each rod. Avoid splashing and dripping. Make sure each rod is completely saturated. Set a timer and neutralize according to the manufacturer's directions.

20 The average time to complete the neutralization process is 10 minutes without the use of a hair dryer.

21 After neutralizing is complete, thoroughly rinse hair with water for about 2 minutes with rods still in the hair. Do not remove rods during rinsing.

22 Gently blot to remove excess water.

23 Remove the rods from the hair, and rinse thoroughly for about 2 minutes.

24 Towel-dry and apply hydrating conditioner. Using a large-tooth comb, distribute the conditioner throughout the hair.



25 Rinse, towel-dry and style as desired.



CAUTION

Hair must not be shampooed for at least 48 hours, otherwise curls will be compromised!

CLEAN-UP AND DISINFECTION

- ☐ Clean and disinfect tools and implements.
- ☐ Clean and disinfect work area.
- ☐ Dispose of single-use items. Place all used linens, towels, and capes in the laundry.
- ☐ Wash your hands.

REVIEW QUESTIONS

- 1 Explain the physical and chemical changes that occur in the hair as a result of chemical texture services.
- 2 What characteristics or condition of the hair and scalp should be analyzed before performing chemical texture services?
- 3 List the similarities and differences between permanent wave, curl reformation, and hair relaxing processes.
- 4 List the chemical products used in permanent waving.
- 5 List at least five permanent waving safety precautions.
- 6 List the chemical products used with thio relaxers.
- 7 List the products used with hydroxide relaxers.
- 8 Explain the difference between base and no-base relaxers.
- 9 What chemical texture service requires a pre-service shampoo? Which services do not?
- 10 List the chemical products used in the curl reformation service.
- 11 Explain the difference between a texturizer and a chemical blowout.
- 12 List at least five chemical hair relaxing safety precautions.

CHAPTER GLOSSARY

acid-balanced waves	p. 597	permanent waves that have a 7.0 or neutral pH; do not require hair-dryer heat
alkaline or cold waves (AL-kuh-line WAYVZ)	p. 596	perms that process at room temperature without heat with a pH range between 9.0 and 9.6
ammonium thioglycolate (ATG) (uh-MOH-nee-um thy-oh-GLY-kuh-layt)	p. 596	main active ingredient or reducing agent in alkaline waves
base control	p. 593	the position of the perm rod in relation to its base section
base direction	p. 593	angle at which the perm rod is positioned on the head; also, the directional pattern in which the hair is wrapped
base relaxers	p. 606	relaxers that require the use of a base or protective cream
base sections	p. 592	subsections of panels into which the hair is divided for perm wrapping; one rod is normally placed on each base section
bookend wrap	p. 589	perm wrap in which an end paper is folded in half over the hair ends
chemical blowout	p. 610	partially straightens the hair with the intent that it will be picked out and cut
chemical hair relaxing	p. 579	the process of rearranging the basic structure of extremely curly hair into a straightened form

chemical texture services (KEM-uh-kul TEKS-chur SER-vicez)	p. 578	hair services that cause a chemical change that permanently alters the natural wave pattern of the hair
croquignole perm wrap (KROH-ken-yohl)	p. 594	rodding from the hair ends to the scalp
curl reformation	p. 579	a soft-curl permanent; combination of a thio relaxer and thio permanent, whereby the hair is wrapped on perm rods; used to make existing curl larger and looser
end wraps	p. 589	end paper; absorbent papers used to protect and control the ends of the hair during perming services
endothermic waves (en-duh-THUR-mik WAYVZ)	p. 596	perm activated by an outside heat source, usually a hood-type dryer
exothermic waves (ek-soh-THUR-mik WAYVZ)	p. 597	perms that create an exothermic chemical reaction that heats the solution and speeds up processing
glyceryl monothioglycolate (GMTG) (GLIS-ur-il mon-oh-THY-oh-GLY-kuh-layt)	p. 596	main active ingredient in true acid and acid-balanced waving lotions
hydroxide relaxers (hy-DRAHKS-yd re-LAX-uhrz)	p. 587	very strong alkalis with a pH over 13; the hydroxide ion is the active ingredient in all hydroxide relaxers
lanthionization (lan-THÉE-ohn-iz-ay-shun)	p. 587	process by which hydroxide relaxers permanently straighten hair; lanthionization breaks the hair's disulfide bonds during processing and converts them to lanthionine bonds when the relaxer is rinsed from the hair
lotion wrap	p. 594	permanent waving wrapping technique in which the waving solution is applied to the section before rodding
neutralization (noo-truh-ly-ZAY-shun)	p. 601	process of stopping the action of a permanent wave solution and hardening the hair in its new form by the application of a chemical solution called the neutralizer
no-base relaxers	p. 607	relaxers that do not require application of a protective base
permanent or chemical waving	p. 578	a process used to chemically restructure natural hair into a different wave pattern
pre-wrap solution	p. 598	usually a type of leave-in conditioner that may be applied to the hair prior to permanent waving to equalize porosity
texturize	p. 610	a process used to semi-straighten extremely curly hair into a more manageable texture and wave pattern
thio relaxers (THY-oh re-LAX-uhrz)	p. 587	relaxers that usually have a pH above 10 and a higher concentration of ATG than is used in permanent waving
true acid waves (TROO AS-ud WAYVZ)	p. 596	perms that have a pH between 4.5 and 7.0 and require heat to speed processing; process more slowly than alkaline waves, and do not usually produce as firm a curl as alkaline waves

18



HAIRCOLORING
AND LIGHTENING

CHAPTER OUTLINE

INTRODUCTION

WHY STUDY HAIRCOLORING
AND LIGHTENING?

IDENTIFY THE CHARACTERISTICS
AND STRUCTURE OF HAIR

UNDERSTAND COLOR THEORY

IDENTIFY HAIRCOLORING
PRODUCTS

UNDERSTAND HAIR LIGHTENERS

UNDERSTAND PROCEDURE
AND APPLICATION TERMS

UNDERSTAND PRODUCT
SELECTION AND APPLICATION

DISCUSS HAIRCOLORING
AND LIGHTENING SAFETY
PRECAUTIONS

PROCEDURES 18-1 TO 18-8

LEARNING OBJECTIVES

After completing this chapter, you should be able to:

LO1

Identify six hair characteristics that are analyzed before performing haircoloring services.

LO2

Explain color theory principles as they apply to hair color services.

LO3

Identify haircolor products and explain their actions on hair.

LO4

Explain the action of lighteners on hair.

LO5

Explain procedure and application terms.

LO6

Explain how haircolor products are selected and applied to hair.

LO7

List haircoloring and lightening safety precautions.

figure 18-1a
Client before haircoloring service.



figure 18-1b
Client after haircoloring service.



Introduction

Men and women have altered their hair color for thousands of years. Early cultures considered colors to be symbols of power and mysticism. This belief led to body painting and haircoloring from vegetable and mineral dyes, as evidenced by chemicals and tools found in Egyptian tombs. In the 1880s, American men had their beards and mustaches dyed in barbershops with coloring products that left the hair with strange iridescent tones or purple hues. These early formulations were made of silver nitrate, gold chloride, gum, and distilled water. Since the first synthetic dyes were developed in 1883, color technology and haircoloring processes have steadily improved in performance and safety.

Haircoloring (HAYR-KUL-ur-ing) is the art and science of changing the color of the hair. **Hair lightening** (HAYR LYT-un-ing) is the partial or total removal of natural pigment or artificial color from the hair. In today's barbershops, many clients expect barbers to be skilled in performing haircoloring and lightening services (**Figure 18-1a** and **18-1b**). Recently, there has been an increasing demand for beard and mustache coloring. As a barber, you need to be able to perform these services with knowledge and skill.

why study HAIRCOLORING AND LIGHTENING?

You should study and have a thorough understanding of haircoloring and lightening because:

- Haircoloring and lightening services offer barbers an opportunity for building a reliable and lucrative clientele. The more knowledge and skill you have, the more services you can offer your clients.
- Knowing when to suggest a haircolor service will help build loyalty with your clients and confidence with your barbering skills.
- Barbers need to understand hair structure, the laws of color, and haircoloring procedures in order to be successful when performing haircoloring and lightening services.
- Haircoloring is a chemical service, and chemicals may cause damage and harm to clients and yourself if not used properly. To avoid these situations, it is important that you understand how to use these chemicals safely on clients.

Identify the Characteristics and Structure of Hair

After reading this section, you will be able to:

- LO 1** Identify six hair characteristics that are analyzed before performing haircoloring services.

The client's hair structure and condition will affect the quality and success of the haircolor service. Six characteristics of the hair that are important

considerations in determining haircoloring options and product selection are elasticity, texture, density, porosity, natural hair color, and contributing pigment.

ELASTICITY

Elasticity is an important factor to consider because it is an indication of the strength of the cortex, including cross-bonds and melanin molecules.

- *Normal* elasticity is indicated by wet hair that can stretch up to 50 percent of its original length and then return to that length without breaking.
- *Low* elasticity is indicated by wet hair that does not return to its original length when stretched.

TEXTURE

The diameter of the individual hair strand determines whether the hair texture is classified as fine, medium, or coarse. Melanin is distributed differently within the different textures:

- Fine hair has melanin granules grouped tightly, so the hair takes color faster and may appear darker.
- Medium-textured hair has an average response time to haircolor products.
- Coarse hair has a larger diameter with loosely grouped melanin granules and may take longer to process ([Figure 18-2](#)).

DENSITY

Density refers to the number of hairs per square inch on the scalp. It ranges from thin to thick and determines the subsection size to use to assure proper coverage of the haircolor or lightener. For example, you may need to part off subsections as thin as 1/8 inch on a client with very thick hair.

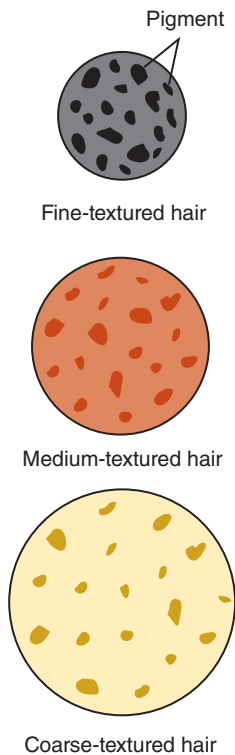
POROSITY

Porosity is the hair's ability to absorb moisture. The porosity level of the hair determines its ability to absorb haircolor products. Porous hair absorbs haircolor products faster and with more intensity and may appear darker than less porous hair. Degrees of porosity are:

- *High porosity*: Hair may not only absorb the color product quickly but also tend to fade quickly due to its inability to hold color pigments. The hair will feel very rough, dry, or may break during a porosity test.
- *Average porosity*: The cuticle is slightly raised and the hair tends to process normally. You should feel a slight roughness to the hair during a porosity test.
- *Low porosity*: The cuticle is tight, making the hair more resistant to chemical penetration, and may require a longer time to penetrate the hair shaft or a higher volume of developer for its strength. The hair will feel smooth and/or hard during a porosity test.

figure 18-2

Melanin distribution according to hair texture.



NATURAL HAIR COLOR

Natural hair color ranges from black to dark brown to red, and from dark blond to lightest blond. These natural color ranges are produced by two types of melanin:

- *Eumelanin* provides natural black and brown pigment (color) to hair.
- *Pheomelanin* provides yellow (blond) and red pigment (color) to hair.

The three factors that determine what natural hair colors look like are the:

- thickness of the hair
- total number and size of pigment granules
- ratio of eumelanin to pheomelanin within the cortex layer

Gray hair is the result of a reduction in the production of melanin pigments. White hair is actually the color of keratin without melanin and therefore does not contain either type of melanin.

CONTRIBUTING PIGMENT (OR “UNDERTONE”)

Contributing pigment (kun-TRIB-yoot-ing PIG-ment) is the pigment that lies under the natural hair color. The foundation of haircoloring is based on modifying this pigment with haircoloring products to create new colors. When lightening a client’s natural hair color, the darker the natural level, the more intense the contributing pigment.

Understand Color Theory

After reading this section, you will be able to:

- LO 2** Explain color theory principles as they apply to haircolor services.

Color is a characteristic of visible light energy. Although the human eye sees only six basic colors, the brain is capable of visualizing combinations of different wavelengths relevant to the three primary and three secondary colors (see Chapter 8). The light rays that are absorbed or reflected by natural hair pigment or artificial pigment, added to the hair, create the colors we see.

THE LAWS OF COLOR

The **laws of color** is a system for understanding color relationships. The laws of color regulate the mixing of dyes and pigments to make other colors. Based in science and adapted to art, the laws of color serve as guidelines for harmonious color mixing. For example, equal parts of red and blue mixed together always make violet. A color wheel is a diagram that presents colors in a specific and sequential order to show the relationship of one color to another.

PRIMARY COLORS

Primary colors (PRY-mayr-ee KUL-urz) are basic or true colors that cannot be created by combining other colors. The three primary colors are yellow, red, and blue (Figure 18-3) and are found naturally in the world, such as yellow dandelions and red roses. All other colors are created by some combination of red, yellow, or blue. Colors with a predominance of blue are “cool-toned” colors and colors that are predominantly red are “warm-toned” colors.

- *Blue* is the strongest and the only cool primary color. Blue brings depth or darkness to any color to which it is added.
- *Red* is the medium primary color. Adding red to blue-based colors makes them appear lighter. Conversely, red added to a yellow color will cause it to appear darker.
- *Yellow* is the weakest of the primary colors and will lighten and brighten other colors.

SECONDARY COLORS

Secondary colors (SEK-un-deh-ree KUL-urz) are created by mixing *equal* amounts of two primary colors. When mixed in equal parts:

- yellow and blue create green
- blue and red create violet
- red and yellow create orange (Figure 18-4)

TERTIARY COLORS

Tertiary colors, also called *quaternary colors*, are created by mixing equal amounts of one primary color with one of its adjacent secondary colors. Tertiary colors are:

- blue-green
- blue-violet
- red-violet
- red-orange
- yellow-green
- yellow-orange (Figure 18-5)

COMPLEMENTARY COLORS

Complementary colors (kahm-pluh-MEN-tur-ee KUL-urz) are primary and secondary colors positioned directly opposite each other on the color wheel. As seen in Figure 18-6, complementary colors include:

- blue and orange
- red and green
- yellow and violet

figure 18-3
Primary colors.

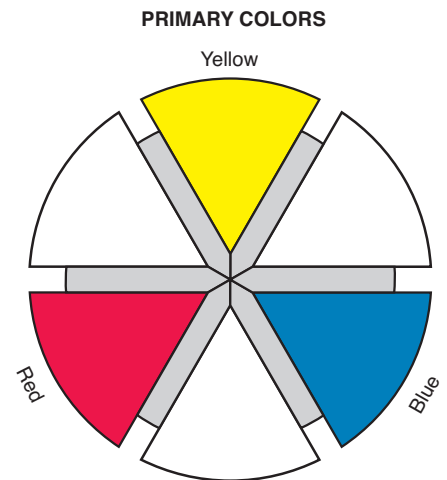


figure 18-4
Secondary colors.

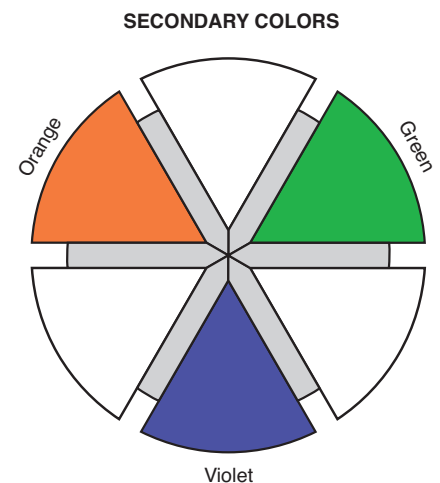


figure 18-5
Tertiary colors.

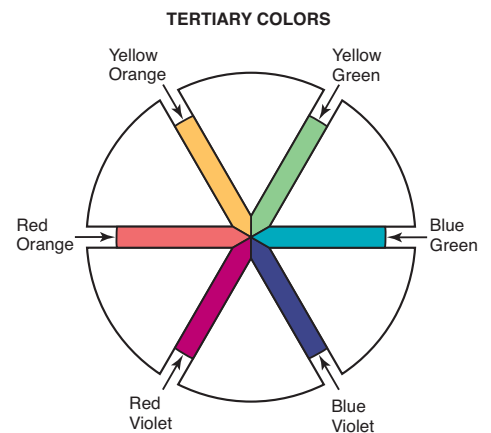
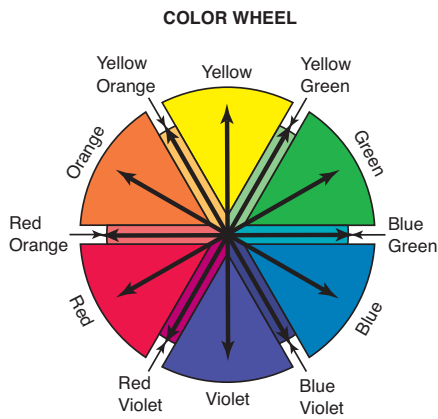


figure 18-6
Complementary colors neutralize each other.



When mixed together, complementary colors “neutralize” each other. For example, when mixed in equal amounts, red and green neutralize each other, creating brown, or a “neutral” tone.

Complementary colors are always composed of a primary and a secondary color, and complementary pairs always consist of all three primary colors. For example, the color wheel shows that the complement of red (a primary color) is green (a secondary color). Green is made up of blue and yellow (both primary colors). So, all three primary colors are represented to varying degrees in the complementary pair of red and green.

HUE AND TONE

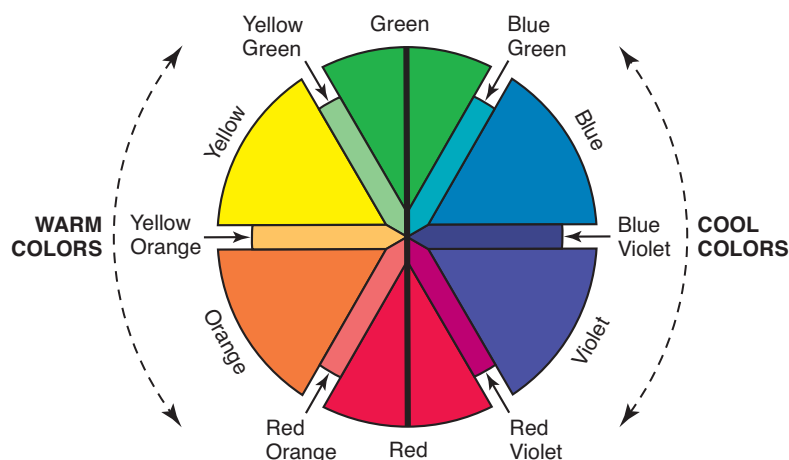
- **Hue** (HYOO) is the basic name of a color, such as red, yellow, or violet. The color wheel is an arrangement of hues that makes the relationships among colors visible.
- **Tone** (TOHN) describes the warmth or coolness of a color (**Figure 18-7**). The warm colors, also known as *highlighting colors*, produce warmer tones because they reflect more light. Warm colors are red, orange, and yellow. The cool colors, also known as *ash or drab colors*, absorb more light and cast cool tones. Cool colors are blue, green, and violet.

LEVEL

Level (LEV-ul) is a unit of measurement used to identify the lightness or darkness of a color. Level is the saturation, density, or concentration of color:

- Colors lighten when mixed with white.
- Colors darken when mixed with black.

figure 18-7
The color wheel divided to represent both warm and cool colors.



The **level system** (LEV-ul SIS-tum) is used to analyze the lightness or darkness of a hair color (**Figure 18-8**). The colors of hair and of haircolor products are arranged on a scale of 1 to 10:

- 1 is the darkest (black)
- 10 is the lightest (blond)

Although the names for the color levels may vary among manufacturers, the level system provides a guide to identify the lightness and darkness of each color level. This information is necessary to determine natural hair color levels and for formulating, matching, and correcting colors.

SATURATION

Saturation, or intensity, refers to the degree of concentration or amount of pigment in the color. It is the strength of a color. For example, a saturated red is very vivid. Any color can be more or less saturated. The more saturated the product, the more dramatic the change in hair color.

BASE COLOR

Artificial haircolors are developed from primary and secondary colors to form “base colors.” The **base color** (BAYS KUL-ur) of a haircoloring product is the predominant tone of a color, which greatly influences the final color result. Examples of base colors used in a haircolor product are as follows:

- a violet base color produces cool results and helps minimize yellow tones
- a blue base color minimizes orange tones
- a red-orange base color creates warm, bright tones in the hair
- a neutral base color tends to soften and balance other colors

IDENTIFYING NATURAL LEVEL AND TONE

Haircoloring results depend on the combination of the:

- client’s natural hair color
- artificial haircolor product that is added to it

Identifying the natural level and tone of the client’s hair will determine which products to use and what the final results will look like. Identifying the natural level of the client’s hair is accomplished by using a manufacturer’s swatch or color ring to match the client’s hair color (**Figure 18-9**).

figure 18-8
The level system.

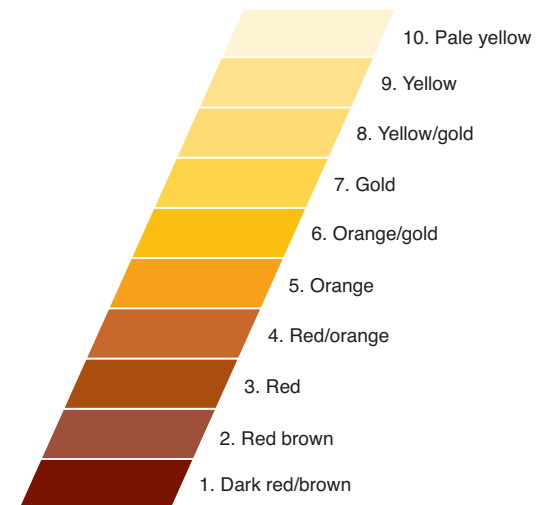


figure 18-9
Manufacturer’s swatches are a useful tool.



Natural Level Mini-Procedure

figure 18-10
Take a ½-inch square section in the crown.



figure 18-11
Hold the color swatch against the hair strand.



- To determine a client's natural level:
1. Part off a ½-inch square section from just under the crown area, comb through, and hold the section so light will filter through (**Figure 18-10**).
 2. Using the natural level swatch or color ring, select a swatch that best matches the hair color.
 3. Move the swatch from the scalp along the entire strand (**Figure 18-11**).
 4. Determine the natural hair color level.

CAUTION

If the hair has been previously colored, use the natural level swatch on the new growth and the manufacturer's color swatches on previously colored sections.

GRAY HAIR

Gray hair is normally associated with aging, although heredity is also a contributing factor. In most cases, the loss of pigment increases as a person ages, resulting in a range of gray tones from blended to solid. The amount of gray in an individual's hair is measured in percentages (**Table 18-1**) and requires special care when formulating haircolor applications.

table 18-1
DETERMINING THE PERCENTAGE OF GRAY HAIR

Percentage of Gray Hair	Characteristics	Level 5 Natural Hair	
30%	More pigmented, or colored, than gray hair		Level 5 natural hair with 30% gray
50%	Even mixture of gray and pigmented hair		Level 5 natural hair with 50% gray
70 to 90%	More gray than pigmented hair; most of remaining pigment is located at the back of the head		Level 5 natural hair with 75% gray
100%	Virtually no pigmented or colored hair; tends to look white		100% gray hair

The challenges and solutions associated with coloring gray hair are discussed later in the chapter (Figure 18-12a and 18-12b).

figure 18-12a
Gray hair presents certain challenges.



Identify Haircoloring Products

After reading this section, you will be able to:

LO3 Identify haircolor products and explain their actions on hair.

Haircoloring products are categorized as nonoxidative and oxidative. The nonoxidizing haircolor products are:

- *temporary haircolor*: washes out with one shampoo
- *semipermanent haircolor*: washes out or fades within a few weeks

The oxidizing haircolor products are:

- *demipermanent*: lasts longer than a semipermanent, but not as long as a permanent haircolor
- *permanent*: stays in the hair until the hair grows out

Nonoxidizing versus oxidizing haircolor products determine a product's colorfastness, or its ability to remain on the hair, and that is determined by the chemical composition and molecular weight of the pigments and dyes within the products (Tables 18-2 and 18-3).

P 18-1 Patch Test *pages 678–679*

P 18-2 Strand Test *pages 680–681*

figure 18-12b
Many haircolor options cover gray successfully.



table 18-2

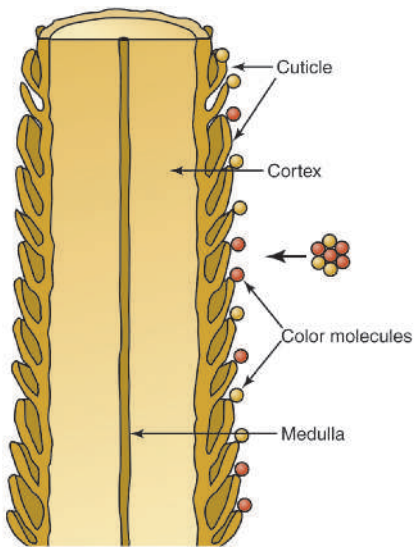
HAIRCOLOR CLASSIFICATIONS AND THEIR USES

Category	Uses
Temporary color	• Create fun, bold results that easily shampoo from the hair (e.g., haircolor sprays used at Halloween) • Neutralize yellow or other unwanted tones (e.g., a blue rinse used on gray hair)
Semipermanent color	• Introduces a client to haircolor services • Adds subtle color results
Demipermanent color	• Blends gray hair • Enhances natural color • Tones pre-lightened hair • Refreshes faded color • Can act as a filler in color correction
Permanent haircolor	• Changes existing haircolor permanently • Covers gray • Creates bright or natural-looking haircolor

CHARACTERISTICS OF HAIRCOLOR CLASSIFICATIONS

Characteristic	Temporary	Semipermanent	Demipermanent	Permanent
Size/weight of dye molecule	Large	Medium	Medium-small	Small
pH	Acid	Slightly alkaline	Moderately alkaline	Alkaline
Type of reaction or change	Physical	Chemical and physical	Chemical and physical	Chemical and physical
How long will it last?	Removed with shampooing	Fades gradually	Some fading, may leave a line of demarcation	Permanent
What does it do?	Deposits on outside of cuticle	Deposits and slightly penetrates cortex	No-lift, deposits and slightly penetrates cortex	Lifts (lightens) and deposits into the cortex
Is it mixed with hydrogen peroxide?	No	No	Yes	Yes
Is a predisposition test required?	One ingredient, aniline derivative (AN-ul-un DUR-ive-it-ive), requires that a predisposition test (also called a <i>patch test</i>) be performed. These tests are required by the Food and Drug Administration (FDA) before the application of any product that contains aniline derivative. Always read the manufacturer’s instructions. If the manufacturer requires a predisposition test, it means that the product contains aniline derivative.			

figure 18-13
Action of temporary haircolor.



TEMPORARY HAIRCOLOR (NON-OXIDATION COLOR)

Temporary colors (TEM-poh-rayr-ee KUL-urz) use color molecules that are the largest found in haircoloring products. The large size of the color molecule prevents penetration beyond the cuticle layer, producing only a coating on the outside of the hair. This coating usually lasts only until the next shampoo (Figure 18-13). The chemical composition of a temporary color is acidic, creating a physical change rather than a chemical change in the hair shaft. Temporary rinses have a pH range of 2.0 to 4.5. If a temporary haircolor has aniline derivative as an ingredient, a predisposition test is required. Be sure to read the manufacturer’s instructions.

Different types of temporary haircolors include:

- *Color rinses* highlight the existing color or add color to the hair and remain on the hair until the next shampoo.
- *Color-enhancing shampoos* are a combination of a color rinse with a shampoo and produce highlights and impart slight color tones to the hair.
- *Crayons* are sticks of coloring compounded with soaps or synthetic waxes and sometimes used to color gray or white hair between hair tint retouches. They can also be used as temporary coloring for mustaches, beards, or sideburns. They are available in several standard colors:

- blond
- light, medium, and dark brown
- black
- auburn
- *Haircolor sprays* are applied to dry hair from aerosol containers. They are usually available in vibrant colors and generally used for special effects.
- *Haircolor mousses and gels* are a combination of a little bit of color and styling mousse or gel in one product.

P 18-3 Temporary Color Rinse pages 682–683

SEMI-PERMANENT HAIRCOLOR (NON-OXIDATION COLOR)

Semipermanent colors are no-lift, deposit-only haircolor products. Semi-permanent pigment molecules are of a lesser molecular weight and size than those of temporary colors. The pigment molecules are small enough to partially penetrate the hair shaft and stain the cuticle layer. However, these molecules are also small enough to leave the hair during shampooing and to fade with each shampoo. Traditional semipermanent colors usually last from six to eight shampoos as the color molecules are shampooed from the hair (Figure 18-14). The chemical composition of semipermanent color is mildly alkaline, causes the cortex to swell, and raises the cuticle to allow some penetration. This chemical composition combines small color molecules, solvents, alkaline swelling agents (mild oxidizer), and surfactants to create a type of color that is known as self-penetrating. Self-penetrating colors tend to make a mild chemical change as well as a physical change. Most semipermanent colors do not contain ammonia and may be used right out of the bottle. Although normally gentle on the hair, semipermanent colors require a patch test (predisposition test) prior to application to prevent the occurrence of product sensitivity or allergic reaction. These tests are required by FDA before the application of any product that contains aniline derivative. Semipermanent haircolor typically falls within the 7.0 to 9.0 pH range. Due to the slight alkalinity of semipermanent color, haircolor services should be followed with a mild, acid-balanced shampoo and conditioning. This process neutralizes any residual alkalinity and helps restore the hair to normal pH levels.

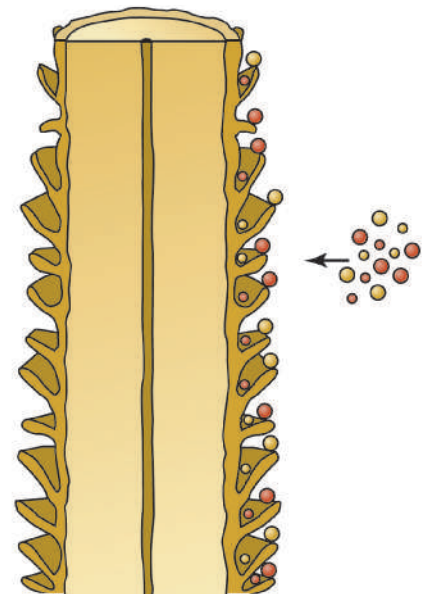
Semipermanent haircolor may be used to:

- cover or blend partially gray hair without affecting its natural color when the hair color and product color are matched correctly
- cover gray hair on a client who has up to 25 percent gray
- enhance or deepen color tones in the hair
- serve as a non-peroxide toner for pre-lightened hair

DEMIPERMANENT HAIRCOLOR (OXIDATION COLOR)

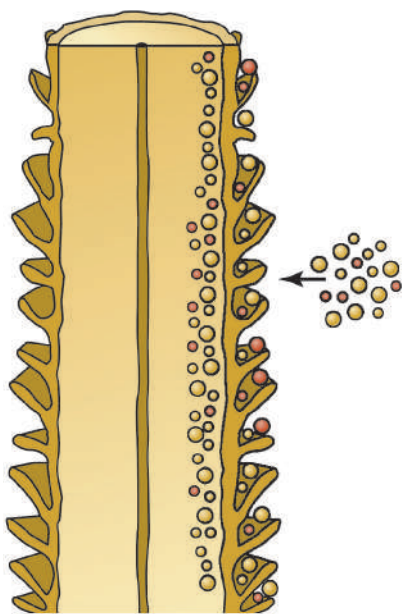
Demipermanent haircolors (DEM-ih PUR-muh-nent HAYR-KUL-urz), also known as *no-lift, deposit-only haircolors*, are longer lasting than semipermanent colors. They are designed to deposit color without lifting (lightening)

figure 18-14
Action of semipermanent haircolor.



? DID YOU KNOW? During the past several decades, some manufacturers labeled their tube color products as semipermanent color, even though a mild developer was required. Since then, most now call these products *demipermanent*.

figure 18-15
Action of demipermanent color.



natural or artificial color in the hair and are a type of oxidation color. They are not used directly out of the bottle but must be mixed with a low-volume developer or activator immediately before use. The agent in the developer causes “oxidation” to occur, which develops the color (**Figure 18-15**). Demipermnants darken the natural hair color when applied. They are available in gel, cream, or liquid forms. Because they contain aniline derivative, a patch test is required before application.

Demipermanent haircolor may be used to:

- get vivid color results
- introduce a client to a color service
- blend or cover up to 50 percent gray
- refresh faded permanent color
- deposit color changes without lightening the hair
- reverse highlights
- perform corrective coloring

P 18-4 Demipermanent Color Applications *pages 684–685*

PERMANENT HAIRCOLOR (OXIDATION COLOR)

Permanent haircolors (PUR-muh-nent HAYR-KUL-urz) are mixed with developers (hydrogen peroxide) and remain in the hair permanently. When the hair grows, a touch-up or retouch application is required to blend the new growth with the previously colored hair. A permanent haircolor is also called a *tint*.

Permanent haircolor products usually contain all of the following:

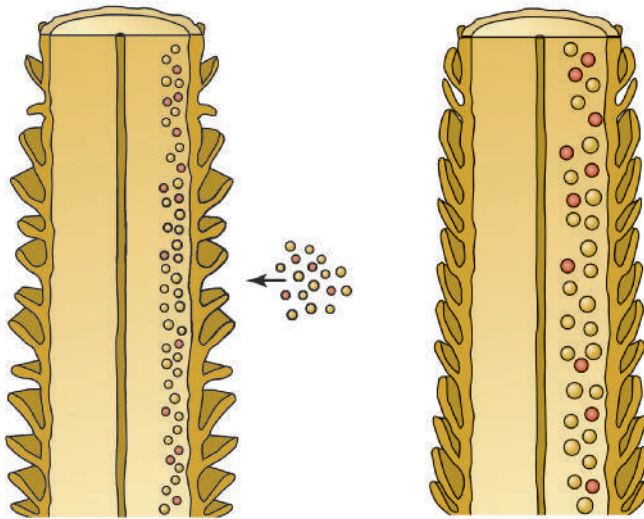
- ammonia
- oxidative tints (color molecules; aniline derivative)
- hydrogen peroxide (developer)

Permanent haircolor products contain aniline derivative, so they require a patch or predisposition test. They can lighten and deposit color in one process. They can lighten natural hair color because they are more alkaline than demipermanent oxidation colors and are usually mixed with a higher-volume developer. The amount of lift is controlled by the pH and concentration of peroxide in the developer. The amount of deposit depends on the amount of color in the product.

Permanent haircolor products are usually mixed with an equal amount of 20-volume peroxide and are capable of lifting one or two levels. When mixed with higher volumes of peroxide, permanent colors can lift up to four levels. Since some manufacturers recommend a 2:1 ratio of developer to haircolor, always read the manufacturer’s directions.

When a permanent haircolor is mixed with a developer and applied to the hair, the cuticle layers will swell and begin to open allowing the undeveloped, small color molecules to enter the cortex. This process usually takes place in the first 10 to 15 minutes after the color is mixed. The molecule (aniline derivative) will then oxidize and grow in size in the remaining processing time, trapping the molecule in the cortex. They are then too big to get out of the cortex (**Figure 18-16**).

figure 18-16
Action of permanent haircolor.



Permanent haircolor products are alkaline and generally range between 9.0 and 10.5 on the pH scale. After processing, the hair is shampooed and as it dries it begins to return to its normal pH. This process causes the cortex to shrink and the cuticle to close, which tends to further trap the color molecules. Except for residual color, permanent haircolor does not wash out during the shampoo process. Eventually the color may fade and may require refreshing.

When new growth occurs, a **line of demarcation** (LYN UV dee-mar-KAY-shun) develops between the new growth and the previously colored hair. This stage requires an application of permanent haircolor to the new growth, known as a *retouch application*.

Permanent haircoloring products are regarded as the best products for covering gray hair. They diffuse natural pigment through the action of lifting and add artificial color to the hair as the color molecules are trapped in the cortex. The gray and non-gray areas blend uniformly with full coverage.

TYPES OF PERMANENT HAIRCOLOR

There are four types of permanent haircolors.

Oxidation Tints

Oxidation tints are also known as aniline derivative tints, penetrating tints, synthetic-organic tints, and amino tints. They can lighten and deposit color in a single process and are available in a wide variety of colors. *Toners* also fall into the category of permanent color and are aniline derivative products of pale, delicate shades designed for use on pre-lightened hair. Most oxidation tints contain aniline derivatives and require a predisposition (patch) test before the service is performed. As long as the hair is of normal strength and in good condition, oxidation tints are compatible with other professional chemical services, such as chemical waves and relaxers. Oxidation tints are sold in bottles, canisters, and tubes, in either a semiliquid or cream form. Tints must be mixed with hydrogen peroxide, which activates the chemical reaction known as *oxidation*. This reaction begins as soon as the

two products are combined, so the mixed tint must be used immediately. Any leftover tint must be discarded since it deteriorates quickly. Timing the application of the tint depends upon the product and the volume of peroxide selected. Consult the manufacturer's directions and your instructor for assistance. A strand test should always be performed to ensure satisfactory results.

Vegetable Tints

Vegetable tints, also known as *natural haircolors*, are haircoloring products made from various plants, such as herbs and flowers. In the past, indigo, chamomile, sage, Egyptian henna, and other plants were used to color the hair. Even though vegetable tints are considered permanent, they are non-oxidation color products because they are not mixed with a developer.

Henna is still used as a professional haircoloring product, but should be used with caution. It has a coating action that can build up with overuse and prevent the penetration of other chemicals. It may penetrate the cortex and attach to the salt bonds, leaving the hair unfit for other professional treatments.

Metallic or Mineral Dyes

Metallic dyes are not professional haircoloring products. Metallic dyes are advertised as *color restorers* or **progressive colors** (pruh-GRES-iv KUL-urz). The metallic ingredients, such as lead acetate or silver nitrate, react with the keratin in the hair, turning it brown. This reaction creates a colored film coating that produces a dull metallic appearance. If a product has silver nitrate in it, it oxidizes over time with oxygen in the air and tarnishes, turning the hair to blackish-gray (much like silverware made of silver does over time if it is not kept polished). Repeated treatments of metallic dyes can damage the hair and can react adversely with many professional chemical services, such as relaxers and chemical waves.

Compound Dyes

Like metallic dyes, compound dyes are not used professionally. Compound dyes are metallic or mineral dyes combined with a vegetable tint. Metallic salts are added to vegetable tints, such as henna, to give the product more staying power and to create different colors. Like metallic dyes, these products can change color, coat the hair, and make hair unfit for other chemical services.

HYDROGEN PEROXIDE DEVELOPERS

A hydrogen peroxide **developer** (dee-VEL-up-ur) is an oxidizing agent that supplies oxygen gas for the development of color molecules when mixed with an oxidative haircolor product. When diluted with water and other substances for use in haircoloring, hydrogen peroxide has a mildly acidic pH of 3.5 to 4.0.

Hydrogen peroxide alone produces a relatively mild lightening of the natural hair color and causes little damage to the hair shaft. A color change occurs in the hair when the oxygen combines with the melanin in the hair. As the oxygen and melanin combine, the peroxide solution begins to diffuse and lighten the melanin within the cortex. The smaller structure and spread-out distribution of the diffused melanin gives the hair a lighter

appearance. This diffused melanin is called *oxymelanin*. When very pale, light shades are desired, however, chemical lighteners must be used.

Hydrogen peroxide (H_2O_2) serves as the main oxidizing agent used in haircoloring. The chemical action of the oxygen with the artificial haircolor molecules in the haircoloring product is called *oxidation*. The small, artificial color molecules (aniline derivative) expand into a larger form because of this chemical action.

In haircoloring, the term *volume* is used to denote the different strengths of hydrogen peroxide (Table 18-4). **Volume** (VAHL-yoom) measures the concentration and strength of hydrogen peroxide:

- The lower the volume, the lesser the lift or lightening achieved.
- The higher the volume, the greater the lifting or lightening action.

Hydrogen peroxide is distributed for use under a variety of names that include:

- developer
- oxidizer
- generator
- catalyst

Regardless of the name used, hydrogen peroxide is available in three forms:

- *Cream peroxides* contain additives such as thickeners, drabbers, conditioners, and acids for stabilization. The thickeners help create a product that tends to stay moist on the hair longer than liquid peroxide, is easy to control, and does not drip during the brush-and-bowl method of application.
- *Liquid hydrogen peroxides* contain a stabilizing acid that brings the pH to between 3.5 and 4.0. They are convenient because they can be used with most of today's bleach and tint formulas.

table 18-4

VOLUMES AND USES OF HYDROGEN PEROXIDE

Volume	Uses
10-Volume	• Used with demipermanent products to deposit color or when less lift/lightening is desired. • Recommended when less lightening is desired.
20-Volume	• The standard volume used with many permanent color products. • Used with permanent color products to cover gray. • Produces up to one or two levels of lift/lightening in one step.
30-Volume	Used with permanent haircolor to achieve up to three levels of lift/lightening in one step.
40-Volume	Used with permanent and high-lift haircolor to achieve up to four levels of lift/lightening in one step.



DID YOU KNOW?

Additives in cream peroxides may dilute the strength of the formula and make it undesirable when full strength is needed.



CAUTION

Plastic bottles of hydrogen peroxide that develop a bulge should be disposed of immediately. The bulge indicates that oxygen atoms have escaped from the hydrogen peroxide and are building up pressure within the container. Open with extreme caution!

- *Dry peroxides* are available in either tablet or powder form and are dissolved in liquid hydrogen peroxide to boost the volume. The availability of liquid and cream peroxides in a variety of volumes has made this product somewhat obsolete.

The safety precautions for using hydrogen peroxide include the following:

- Use clean implements when measuring, using, and storing hydrogen peroxide. Even a small amount of dirt or impurities can cause hydrogen peroxide to deteriorate.
- Never measure the needed amount of hydrogen peroxide by pouring it into the lid of another product. The residue will cause the product in the container to oxidize as it sits on the shelf and render it unusable.
- Do not allow hydrogen peroxide formulations to come in contact with metal. Metal causes the oxidation process to occur too quickly to allow proper color development.
- Avoid breathing in vapors caused by mixing hydrogen peroxide and haircolor products.
- A hydrogen peroxide volume of 20 or more can cause skin irritations, chemical burns, and hair damage.
- Keep the cap closed securely on the hydrogen peroxide at all times when not in use. Overexposure to the air will affect the strength.

ACTIVATORS

An **activator** (AK-tih-vay-ter) is an oxidizer, consisting of powdered persulfate salts, that is added to haircolors, lighteners, or hydrogen peroxide to increase the chemical action of the product. This addition results in an increased lifting power, which is controlled by the number of activators that are added to the product. Generally, up to three activators can be used for on-the-scalp applications and up to four for off-the-scalp processes; however, since formulas and strengths will vary from brand to brand, always refer to the manufacturer's directions for mixing instructions. Activators are also known as boosters, protinators, and accelerators.

P 18-5 Single-Process Permanent Color Applications: Virgin and Retouch *pages 686–688*

Understand Hair Lighteners

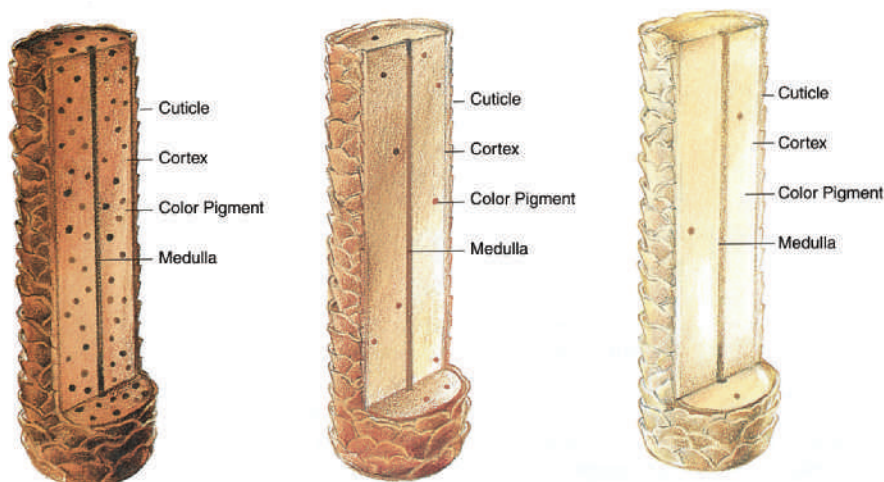
After reading this section, you will be able to:



Explain the action of lighteners on hair.

Lighteners (LYT-un-urz) are chemical compounds that lighten hair by dispersing, dissolving, and decolorizing the natural hair pigment (melanin) (**Figure 18-17**), which is accomplished by a mixture of bleach

figure 18-17
Hair lighteners diffuse pigment.



and hydrogen peroxide. When mixed, the pH of lighteners is around 10.0. As soon as hydrogen peroxide is mixed into the lightener formula, it begins to release oxygen.

The hair pigment goes through different stages of color as it lightens. The amount of change depends on:

- how much natural pigment, or melanin, the hair has
- the strength of the lightener
- the length of time the lightener is on the hair

During this decolorization process, natural hair may go through many stages of lightening from the darkest to the lightest: natural black hair can lighten through the brown and/or red stages to orange, gold, yellow, and finally to pale yellow (**Figure 18-18**). Hair lighteners are used to:

- create blond shades that are not possible with permanent haircolor
- pre-lighten the hair to prepare it for the application of a toner or tint (double-process application)
- lighten the hair to a particular shade or stage
- brighten and lighten an existing shade
- lighten only certain areas of the hair
- lighten naturally dark hair
- lighten hair without depositing color

TYPES OF LIGHTENERS

Lighteners are available in three forms: cream, powder, and oil. Cream and oil lighteners are considered **on-the-scalp lighteners** (AWN-THE-SKALP LYT-un-urz) and powder lighteners are

?
DID YOU KNOW?

It is important to note that the purpose of *lighteners* is to dissolve or remove the hair's natural color pigment *melanin*. It is not the purpose of lighteners to remove artificial haircoloring products, such as tints, from the hair. To remove unwanted *artificial haircoloring* products from the hair, use *tint or dye removers*.

!
CAUTION

Do not use a lightening product if the scalp shows any signs of sensitivity, cuts, or abrasion.

figure 18-18
Ten degrees of decolorization.



off-the-scalp lighteners (AWF-THE-SKALP LY-Tun-urz). That said, some newer powder lighteners may be used on the scalp, so always refer to the manufacturer's directions before purchase or application. Each type of lightener has unique abilities, chemical compositions, and formulation procedures.

- *Cream lighteners* are the most popular type of on-the-scalp lightener. They contain conditioning agents, bluing agents, and thickeners, which makes them easy to apply, and will not run, drip, or dry out. Cream lighteners provide the following benefits:
 - The conditioning agents give some protection to the hair.
 - The bluing agent helps drab undesirable red and gold tones.
 - The thickener provides control during application and prevents overlap.
- *Powder lighteners*, also called *paste*, *speed*, or *quick lighteners* contain oxygen-releasing boosters and substances for quick and strong lightening. These lighteners will stay in place and not run or drip, but do not contain conditioning agents and tend to dry out quickly. Because they are stronger than oil or cream lighteners, they may also be too strong to use directly on the scalp. Be sure to read the directions to see if it is safe to use powder lightener for a virgin lightener or lightener retouch.
- *Oil lighteners* are not as popular as cream or powder lighteners nowadays. They are usually mixtures of hydrogen peroxide with sulfonated oil. As on-the-scalp lighteners, they are the mildest form of lightener and may be used when only one or two levels of lift are desired. There are two types of oil lighteners:
 - *Color oil lighteners* add temporary color as they lighten:
 - *Gold*: lightens and adds golden to reddish tones depending on the base color of the hair
 - *Silver*: lightens and adds silvery highlights to gray or white hair and minimizes red and gold tones in other shades
 - *Red*: lightens and adds red highlights
 - *Drab*: lightens and adds ash highlights, and tones down or reduces red and gold tones
 - *Neutral oil lighteners* remove pigment without adding color tone and may be used to pre-soften hair for a tint application.

CONTRIBUTION OF UNDERLYING PIGMENT

It is essential to lighten the hair to the correct stage of lightness needed, because the pigment that remains in the hair will impact the final result of the hair lightening and coloring process. For example, let's say you need to lighten to the pale yellow stage, but only lighten to the yellow or gold stage. The tint or toner you have selected has a blue base. If you were to apply this blue-based tint to the yellow/gold hair, the hair will turn green because you only lightened to the yellow/gold stage, and blue added to yellow (think color theory) makes green. Use [Figure 18-19](#) as a guide to determine the contributing pigment or undertones of color at various hair color levels.

figure 18-19
Contributing pigment (undertones).



TONERS

A toner is a permanent haircoloring product that is applied to pre-lightened hair for the purpose of achieving the desired color or tones in the hair or to neutralize unwanted undertones (contributing pigment). For example, brassy red tones that remain in the hair after lightening can be “neutralized” with a green-based toner if they are at the same level of lightness. Toners only add or deposit pigment into the hair shaft; they do not lighten the hair. Because toners are applied only to pre-lightened hair, they typically come in pale, delicate, blond shades. Aniline derivative is an ingredient in toners, so a patch test is required 24 to 48 hours before application. Toners usually have a very different color in the bottle than the final shade they produce. They may appear to be purple, blue, orange, or pink in the bottle. As toner color oxidizes, it goes through several visual color changes; therefore, a preliminary strand test should be done to determine the processing time required for a desired shade. Toners are used in double-process applications:

1. the lightener application is the first process
2. the toner application is the second process

After the hair goes through the desired stages of lightening, the color left in the hair is known as its *contributing color*, which is usually the darkest degree of contributing pigment that remains after the lightening process. Achieving the correct lightening stage is necessary for proper toner development and to achieve even color. Toner manufacturers provide recommendations about the contributing color needed to achieve various color results. As a general rule, the lighter the desired color, the lighter the foundation must be. It is important to follow the guide closely.



CAUTION

Advise clients that lightening dark hair to a pale blond tone can be very damaging to the hair.

Overlightened hair will absorb or *grab* the base color of the toner, while underlightened hair will appear to have more red, yellow, or orange than the intended color. While color manufacturers produce toners to complement their color lines, toning is also a process that can be performed using a semipermanent, demipermanent, or permanent haircolor as a “toner” to achieve the desired result. Read the manufacturer’s directions carefully for appropriate color selection, mixing, and application instructions. Toner formulations and color ranges vary with each manufacturer. Using products from the same product line is recommended.

TINT OR DYE REMOVERS

The removal of artificial haircoloring products, such as a tint, is sometimes desired if:

- the client wants to change to a lighter shade
- a coloring mistake has been made
- the hair has processed too dark due to an overporous condition

Dye removers (DYE ree-MOOV-urz) are also known as *color or tint removers*.

There are two basic types of products available to remove artificial pigment, such as tints and toners, from the hair:

- *Oil-base dye removers* lift trapped color pigments, stain, or buildup from cuticle layers, do not create structural changes in the hair shaft or pigment (natural or artificial) of the hair, and will not make drastic changes in the level of color.
- *Tint or dye solvents* produce strong lightening effects on melanin and diffuse artificial pigment within the cortex, are nonallergenic, and do not require a predisposition test because they do not contain aniline derivative. As these are strong and fast-acting, follow the manufacturer’s directions carefully.

FILLERS

Fillers (FIL-urz) are designed to correct excessive porosity and/or to create a color base in the hair by penetrating the cuticle and filling in empty, small holes or pockets in the cortex. They help even out porosity in the hair shaft that can cause uneven color deposit or lightening. There are two types of fillers:

- Conditioner fillers are available in protein and nonprotein and in gel, cream, and liquid forms.
- **Color fillers** (KUL-ur FIL-urz) are dual-purpose haircoloring products that are able to create a color base and equalize excessive porosity in one application. They are available in clear, neutral, and a variety of colors:
 - A clear filler is designed to correct porosity without affecting color and does not deposit a color base.

- Neutral fillers (a balance of all three primary colors) have minimal saturation and color correction abilities, but have power to equalize porosity.

Hair with uneven color distribution or pre-lightened hair may benefit from a color filler application before applying a toner or tint. Color fillers can be used to:

- deposit color to faded hair shafts and ends
- help hair hold color
- help ensure a uniform color from the scalp to the hair ends
- prevent color streaking
- prevent off-color results
- prevent dullness
- facilitate more uniform color in tinting the hair back to its natural shade

Color fillers use certified colors as pigments and are safe to use without a predisposition test if they contain no aniline derivative, may be used directly from the container and applied to the hair before tinting, and should match the same basic shade as the toner or tint to be used.

SKIN STAIN REMOVERS

Generally, soap and water will remove most tint stains from the skin. Stain removers are commercially prepared solutions that are designed for this purpose. When soap and water is not capable of removing haircolor from the skin, use one of the following methods:

1. Dampen a piece of cotton with the leftover tint. Use a rotary movement to cover the stained areas and follow with a damp towel. Apply a small amount of face cream and wipe clean.
2. Use a prepared stain remover.

Understand Procedure and Application Terms

After reading this section, you will be able to:

LO5 Explain procedure and application terms.

Successful haircoloring usually requires a series of steps to accomplish the desired end result. Due to the wide range of haircoloring products, application methods, and procedures, it is important to have a clear understanding of the terms used in haircoloring processes. Review the following section to become familiar with procedure and application terms used in haircoloring and lightening services (see [Table 18-5](#)).

PROCEDURE AND APPLICATION TERMS

Patch or predisposition test	An individual's reaction to aniline derivative tints can be unpredictable. Some clients may show an immediate sensitivity, while others may suddenly develop an allergy to the product after years of use. To identify a client who has a sensitivity to aniline derivatives, the U.S. Federal Food, Drug, and Cosmetic Act prescribes that a patch test (PACH TEST), also known as a <i>predisposition test</i> , be given 24 to 48 hours prior to each application of an aniline derivative tint or toner (refer to Procedure 18-1).
Strand test	A strand test (STRAND TEST) is performed for color applications to determine: • how the hair will react to the haircolor product • how long it will take to process • what the final outcome will look like After the results of the patch test, the strand test is the next step in performing a haircolor service (refer to Procedure 18-2). A strand test is a smart and safe thing to do, but, unlike patch tests, it is not required by law.
Soap cap	A soap cap (SOHP KAP) is a combination of equal parts of a permanent haircolor, hydrogen peroxide, and shampoo that is applied like a regular shampoo. An example of a soap cap mixture is 1 ounce of permanent haircolor, 1 ounce of developer, and 1 ounce of shampoo. Soap caps can be used to: • brighten existing color • reduce unwanted yellow tones in gray hair • blend lines of demarcation when a retouch application does not quite match the former color application
Tint back	<i>Tint back</i> is the term used to describe the process of coloring hair back to its natural color (or close to its natural color). It is important to keep in mind that previously processed hair will be more porous and, therefore, may process darker than intended. In some cases, a filler is required to even out the hair's porosity or to achieve accurate color correction. A demipermanent haircoloring product is usually an effective choice because it is a deposit-only color with minimal oxidation.
Virgin application	A virgin application (VUR-jin ap-lih-KAY-shun) is the application of haircolor to hair that has not been previously colored. Hair that is in a "virgin" state is usually healthy hair that does not have any chemical damage. A virgin application also indicates that the haircoloring product will be applied to the entire hair strand versus the new growth only.
Retouch application	When permanent haircolor or lighteners are used, new hair growth will become obvious after a few weeks. The new growth, or regrowth, is that section of the hair shaft between the scalp and the hair that has been previously colored or lightened. This growth creates a <i>line of demarcation</i> between the natural color of the hair and the previously colored or lightened hair that requires blending by applying color or lightener to the new hair growth. The term retouch application (ree-TUCH ap-lih-KAY-shun) is used to describe this blending process.
Single-process haircoloring	Single-process haircoloring (SING-gul-PRAH-ses HAYR KUL-ur-ing) is a process that lightens (or lifts) and deposits color, or only deposits color in the hair in one, single application. Examples of single-process coloring are: • virgin tint applications • tint retouch applications Single-process haircoloring is also known as: • single-application coloring • one-step coloring • one-step tinting • single-application tinting

Double-process haircoloring

Double-process haircoloring (DUB-ul PRAH-ses HAYR-KUL-ur-ing) requires two separate and distinct applications to achieve the desired lighter hair color:

1. The hair is lightened.
2. A depositing color is applied.

This process allows the practitioner to independently control the lightening and coloring actions. Double-process haircoloring is also known as:

- double-application coloring
- two-step coloring
- two-step tinting
- double-application tinting

Double-process haircoloring may include the use of:

- lighteners and toners
- lighteners and tints
- pre-softeners and tints
- fillers and tints

Pre-softening

Pre-softening (pree-SOF-en-ing) is the process of treating gray or other resistant hair to facilitate better color penetration. Pre-softening swells and opens the cuticle. It can be accomplished with a mixture of 1 ounce of 20-volume peroxide and 8 drops of 28 percent ammonia water or with an oil or cream bleach product.

Highlighting

Highlighting (HY-lyt-ing) is the process of coloring some of the hair strands lighter than the natural or artificial color to add an illusion of sheen and depth. Forms of highlighting include:

- cap frost
- foil frost
- tipping
- streaking
- painting/free-form

Lowlighting

Lowlighting or *reverse highlighting* is the process of coloring strands or sections of the hair darker than the natural or artificial color. Contrasting dark areas appear to recede and make detail less visible to the eye. Forms of lowlighting include:

- cap frost
- foil frost
- tipping
- streaking
- painting/free-form

Cap technique

The **cap technique** involves pulling strands of hair through the holes of a perforated cap with a hook. A lightener, haircolor, or both (lightener and haircolor) are applied to only these hair strands, giving a natural, streaked look. The number of strands pulled through the cap determines the degree of highlighting or lowlighting that is achieved.

Foil technique

The **foil technique** (FOYL tek-NEEK) involves slicing or weaving out sections of hair to be placed on a piece of foil. The color or lightening product is usually brushed onto the hair section within the foil, after which the foil is folded and sealed for processing.

Free-form technique

The **free-form technique** (FREE-form tek-NEEK), or *balyage* (also spelled baliage), is the process of painting a lightener or color directly onto clean, styled hair. The effects can be subtle or dramatic, depending on the type of product (color or lightener) and the amount of hair that it is applied to.

RECORD KEEPING

Before performing a haircoloring service, a client record card should be completed for each client (Figure 18-20). The client record card is used to log all information pertaining to the haircoloring service. In addition to the

figure 18-20

Haircolor record card.

HAIRCOLOR RECORD					
Name _____			Tel. _____		
Address _____			City _____		
Patch Test: ☐ Negative ☐ Positive			Date _____		
Eye Color _____			Skin Tone _____		
DESCRIPTION OF HAIR					
Form	Length	Texture	Density	Porosity	
☐ straight	☐ short	☐ coarse	☐ sparse	☐ very porous	☐ resistant
☐ wavy	☐ medium	☐ medium	☐ moderate	☐ porous	☐ very resistant
☐ curly	☐ long	☐ fine	☐ thick	☐ normal	☐ perm. waved
Natural hair color _____					
	level (1-10)	Tone (Warm, Cool, etc.)	Intensity (Mild, Medium, Strong)		
Scalp Condition					
☐ normal	☐ dry	☐ oily	☐ sensitive		
Condition					
☐ normal	☐ dry	☐ oily	☐ faded	☐ streaked (uneven)	
% unpigmented _____		Distribution of unpigmented _____			
Previously lightened with _____			for _____ (time)		
Previously tinted with _____			for _____ (time)		
☐ original hair sample enclosed		☐ original hair sample not enclosed			
Desired hair color _____					
	level (1-10)	Tone (Warm, Cool, etc.)	Intensity (Mild, Medium, Strong)		
CORRECTIVE TREATMENTS					
Color filler used _____			Conditioning treatments with _____		
HAIR TINTING PROCESS					
whole head _____		retouch inches (cm) _____		shade desired _____	
formula: (color/lightener) _____			application technique _____		
Results:	☐ good	☐ poor	☐ too light	☐ too dark	☐ streaked
Comments: _____					

Date	Operator	Price	Date	Operator	Price
_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____

client's contact information, the record card should be descriptive enough that it provides pre-service and post-service data about the client's hair. For example, key information items should include:

- characteristics of the hair's condition
- scalp condition
- haircolor history
- any corrective treatments
- results of the haircoloring process

This information can be used for future visits as a basis for other services and should be maintained from one visit to the next.

A release statement form should be used when the client's hair is in a questionable condition that may not withstand chemical processes and treatments. See **Figure 18-21** for a sample barber school release form. To some degree, the release statement is designed to protect the school or shop owner from responsibility for accidents and damages and is a requirement of most malpractice insurance. It should be noted, however, that a release statement is not a legally binding contract and will not fully protect the barber or the shop from liability.

figure 18-21
Sample school release form.

RELEASE FORM
I, the undersigned, _____ (name)
residing at _____ (street, address)
_____ (city, state and zip)
about to receive services in the Clinical Department of _____
and having been advised that the services shall be performed by either students, graduate students, and/or instructors of the school, in consideration of the nominal charge for such services, hereby release the school, its students, graduate students, instructors, agents, representatives, and/or employees, from any and all claims arising out of and in any way connected with the performance of these services.
The Proprietor Is Not Responsible for Personal Property
Signed _____
Date _____
Witnessed _____
THIS RELEASE FORM MUST BE SIGNED BY THE PARENT OR GUARDIAN IF THE CLIENT BEING SERVED IS UNDER 18 YEARS OF AGE.

CLIENT CONSULTATION

A thorough client consultation is the first step in a haircoloring service. Consultations should be held in a well-lit room that provides either a strong natural light or incandescent lighting. Fluorescent lighting is not suitable for judging existing hair colors.

Use the following as a guide to perform a haircoloring service consultation:

1. Drape the client.
2. Have the client fill out a client record card.
3. Perform a hair and scalp analysis and write the results on the record card. Use color swatches to determine client's natural level (**Figure 18-22**).
4. Ask the client leading questions about the desired end result to determine the:
 - a. preferred color
 - b. product (temporary, permanent, etc.)
 - c. method (all-over color, retouch, highlights, etc.)
5. Show examples of appropriate colors and make a determination with the client (**Figure 18-23**). Ask if they have any pictures of a hair color they like or don't like.
6. Review with the client:
 - a. the procedure
 - b. application technique
 - c. maintenance needed
 - d. cost of the service
7. Perform a patch test if the haircolor product requires it.
8. Gain the client's approval and begin the service.
9. Record end results on the client record card.

figure 18-22

Use color swatches to determine the natural level.



figure 18-23

Discuss appropriate colors with the client.



Understand Product Selection and Application

After reading this section, you will be able to:

- LO 6** Explain how haircolor products are selected and applied to hair.

Given the many choices in haircoloring formulations and applications, it is important that you provide the client with the appropriate product and follow the correct application method. Use the following as a guide for haircoloring product selection and application.

TEMPORARY COLOR RINSES

Temporary color rinses may be used to give clients a preview of how a color change will look. They are also an option for clients who want to highlight the color of their hair or add slight color to gray hair. These rinses wash out when shampooed and are available in a variety of color shades. Temporary rinses are easily and quickly applied at the shampoo bowl and can serve as an introduction to other, longer lasting color services. Temporary color rinses can be used to:

- bring out highlights
- temporarily restore faded hair color to its natural shade
- neutralize yellow tones in white or gray hair
- tone down overlightened hair

Perform a preliminary strand test to determine proper color selection.

SEMIPERMANENT HAIRCOLOR

Semipermanent haircolor (sem-ee-PUR-muh-nent HAYR-KUL-ur) products are appropriate for the client who may want more color change than is available with a temporary rinse, but who is hesitant about a permanent color change and its related maintenance. A semipermanent color fills the gap between temporary color rinses and permanent haircolor without replacing either of them.

Since semipermanent products are deposit-only colors, the final outcome will depend on the:

- hair's original color and texture
- color that is applied
- length of development time

These haircoloring products are available in liquid and cream forms in a variety of colors. Some formulations are specifically designed in blue-gray or silver-gray hues to brighten or blend gray color tones.

Characteristics of Semipermanent Colors

The basic characteristics of semipermanent haircolors that influence the decision to choose this color product over another are:

- Semipermanent tints do not require the use of hydrogen peroxide.
- The color is self-penetrating to the extent that it stains the cuticle and deposits color molecules into the cortex.
- The color is applied the same way for each application.
- Hair will usually return to its natural color after six to eight shampoos, provided a mild, non-stripping shampoo is used.
- Retouching is eliminated.
- Semipermanent colors contain aniline derivative, so a patch test is required.
- Some semipermanent haircolors require pre-shampooing; others do not. Always follow the manufacturer's instructions.

Selecting Semipermanent Color

The addition of artificial color to the natural pigment in the hair shafts creates a darker color. When using a color chart to determine the level and shade of semipermanent color to use, consider the natural color to represent half of the formula. Use the following to select the correct color to perform the strand test:

- The use of ash or cool shades will create a color that appears darker than if a warm shade is applied.
- Warm colors appear shinier due to the reflection of light.
- For clients with up to 25 percent gray, select a shade that matches the client's natural hair color.

Special Problems

Some semipermanent haircolor products have a tendency to build up on the hair shaft with repeated applications. If this happens:

1. apply the semipermanent color to the new growth only
2. process until the desired color shade develops
3. wet the hair with warm water
4. blend the color through the hair for a few minutes
5. rinse or shampoo the color from the hair, following the manufacturer's instructions

DEMIPERMANENT HAIRCOLOR

Since demipermanent color is considered to be deposit-only color, the same procedures used for the application of a semipermanent haircolor product can be used. The difference between a semipermanent and a demipermanent color is that you mix demipermanent color with a low-volume developer or an activator immediately before applying it to the client's hair. Follow the manufacturer's guidelines for application, color selection, and processing time.

PERMANENT HAIRCOLOR

Professional permanent haircolor products contain aniline derivative color molecules and are mixed with developer. These penetrating tints are available in liquid, cream, and gel forms and are used as either single-process or double-process tints.

Single-Process Tints

Single-process tints provide a simple method of haircoloring. In one application, the hair can be permanently colored without pre-shampooing, pre-softening, or **pre-lightening** (pree-LYT-tin-ing). A single-application tint is applied on dry hair. If the hair is extremely oily or dirty and a shampoo is necessary, it must be dried before applying the tint. Single-process tints contain aniline derivative molecules and an alkaline agent and are formulated for use with 20-volume hydrogen peroxide. When other volumes of peroxide are used, the color results change. The choice of colors available vary from deepest black to lightest blond. Some characteristics of single-process tints are that they:

- save time by eliminating pre-lightening
- color the hair lighter or darker than the client's natural color
- blend in gray or white hair to match the client's natural hair color
- tone down streaks, off-shades, discoloration, and faded hair ends

Color Selection of Single-Process Tints

The porosity of the hair is one of the most important characteristics to consider when choosing haircolor tint shades. Use the following guide for choosing the level of color when tinting darker:

- *Normal porosity*: half level lighter than desired color
- *Slightly porous*: one level lighter than desired color
- *Very porous*: one to two levels lighter than desired color

General rules for single-process color selection for gray hair:

- To match the natural color of hair and to cover gray, select the color closest to the natural shade.
- To brighten or lighten hair color and to cover gray, select a shade lighter than the natural color. The selected tint must contain enough color to produce the desired shade on gray hair.
- To darken the hair and cover gray, select a color darker than the natural hair color.
- Study the manufacturer's color chart for correct color selections.
- If a vibrant color is desired, add a natural shade to it in order to ensure better coverage.

DOUBLE-PROCESS HAIRCOLORING

Double-process haircoloring begins with hair lightening, followed by either a tint or a toner.



DID YOU KNOW?

Skin tones change with age. The natural color of the client's hair, which harmonized with the skin coloring at the age of 20 or 30, may seem harsh and unbecoming at the age of 40 or 50. For clients in this age group, keep to the lighter shades of color.

This double process requires two separate steps as discussed in this section and demonstrated in Procedure 18-6.

P 18-6 Double-Process Haircoloring pages 689–693

LIGHTENERS

Lightening creates a new color foundation that is lighter than the client's natural hair color. This new color foundation may be the finished result or it may be the first step of a double-process application.

To achieve the desired shade of lightness, consideration must be given to the:

- existing hair color
- processing and development time
- resulting porosity
- color selection

Depending on the manufacturer's directions, hair lighteners can be used to:

- lighten the entire head of hair
- lighten the hair to a particular shade or stage
- brighten and lighten the existing shade
- paint, streak, or frost certain sections of the hair
- remove undesirable casts and off-shades
- correct dark streaks or spots in hair that has already been lightened

Selection of Lighteners

Together with the manufacturer's directions, be guided by the following general rules when choosing a lightening product:

- *Cream lighteners* can be used on the scalp when performing virgin or retouch lightening services. They offer some protection to the hair, are controllable during application, and can be used to drab undesirable red and gold tones. For increased strength, up to three activators can be added for on-the-scalp applications and up to four activators for off-the-scalp processes.
- *Powder lighteners* are strong enough to lighten the hair through several stages of lightening and may be too strong to use directly on the scalp—read and follow the manufacturer's directions carefully.
- *Oil lighteners* are the mildest form of lighteners and may be used when only one or two levels of lift are desired.

Lightener Retouch

Lightener retouch is the term used when a lightener is applied only to the new hair growth to match the rest of the lightened hair. The client's record card should be used as a guide to determine the:

- lightener that was previously used
- processing time required for the desired stage of lightness to develop

Cream lightener is often used for a lightener retouch because it helps prevent overlapping on the previously lightened hair. Powdered lighteners may be used if the manufacturer's directions state that it is safe to use on the scalp. When retouching, the lightener is applied to the new growth only. If a lighter or different level is desired overall, wait until the new growth is almost light enough or has developed fully. Then distribute the remainder of the lightener through the hair shaft. One to five minutes should be ample time to create a lighter-level effect.

TONERS

Toners (TOHN-urz) have the same chemical ingredients as permanent hair-color tints, except they contain less amounts of color, which gives toners pale, delicate shades of color for depositing in pre-lightened hair.

Color Selection of Toners

Pastel colors, such as silver, platinum blond, and beige-blond, are popular toners for lighter blond colors. Gray hair and skin tone changes that accompany advancing years may benefit from light, silver tones. When extremely pale toner shades such as very light silver, platinum, or beige are desired, the hair must be pre-lightened to pale yellow or almost white.

Toner Retouch

A toner retouch must be given the same careful consideration as you would give a double-process tint retouch application. The new growth must be pre-lightened to the same degree of lightness achieved in the previous application. The lightener is applied to the new growth only. To avoid damage to the hair, be careful not to overlap the lightener on previously lightened hair. After the lightening process has been completed, follow the manufacturer's instructions for toner application:

- Some toners are applied to the entire length of the hair at one time.
- Some toners are first applied to the new growth area to process, and then applied briefly to the length of the remaining hair.

Toner Suggestions and Reminders

Toners are completely dependent on the proper preliminary lightening treatment, which must leave the hair light and porous enough to receive the pale toner shades. Semipermanent and demipermanent color can be used after lighteners to achieve specific tones and colors. Strand tests are vital in double-process applications. A complete explanation of the possible outcome should be discussed with the client. It is always possible that the hair cannot be decolorized sufficiently for the color choice without resulting in serious damage to the hair. Gold or red pigments remaining in the hair after lightening indicate underlightening. Ash tones indicate overlightening—when this happens, the shade of toner should be chosen to neutralize the unwanted tones.

SPECIAL-EFFECTS HAIRCOLORING AND LIGHTENING

Special-effects haircoloring refers to any technique that involves the partial lightening or coloring of the hair. One way of creating special effects is to strategically place light and dark colors in the hair. Highlighting is the process of lightening or coloring some of the hair strands lighter than the natural color. Lowlighting, or reverse highlighting, is the process of coloring strands or sections of the hair darker than the natural color. Both of these techniques can be used to create special-effects haircoloring and may consist of an overall dramatic dimensional change or something more subtle. Consult your client about where they would like to see the change in color. The three most frequently used techniques for creating highlights or lowlights are:

- cap frost
- foil frost
- balyage, free-form, or painting

P 18-7 Special Effects *pages 694–698*

SPECIAL PROBLEMS AND CORRECTIVE HAIRCOLOR

Each haircoloring or lightening service has the potential to create unique problems. Some problems can be avoided by performing preliminary strand tests, but others can be the result of unique properties within the client's hair structure that are unforeseen. Most haircoloring and lightening problems can be resolved with:

- a calm approach
- an accurate assessment of the problem
- knowledge to correct the situation

figure 18-24

Many people choose to cover or blend gray hair.



GRAY HAIR CHALLENGES

Gray, white, or salt-and-pepper hair shades have characteristics that can present unique color challenges (**Figure 18-24**). Since both gray and white hair contain little melanin within the cortex, a large number of coloring services are performed with the intent to cover or enhance the color. Depending on the amount of gray, the hair may have a yellowish cast or process differently from one strand to another. Some gray hair also tend to be resistant to chemical processes and may require pre-softening before a service.

Gray Hair with a Yellowish Cast

Gray, white, and salt-and-pepper hair with a yellowish cast can be treated with violet-based colors that range from highlighting shampoos and temporary rinses to lighteners. The longevity of the product used will depend on the client's desired result and the options offered

by the barber. If lightening and coloring services are not typically offered in the barbershop, it is highly recommended that, at a minimum, high-lighting shampoos or temporary rinses with violet bases be available to clients.

Determining the Percentage of Gray

Since most people retain some dark hair as they turn gray, the hair must be analyzed for level, hue, and percentage of gray before the appropriate product selection can be made. Gray hair may be evenly distributed or isolated in various sections of the head, such as the temple areas.

Formulating Haircolor for Naturally Gray Hair

Gray hair will usually accept the level of the color applied. Because there is no melanin in the hair, gray hair may appear lighter after haircolor is applied.

When a client has 80 to 100 percent gray, lighter haircolors are usually more flattering than darker shades. The client's skin tone, eye color, and personal preference will determine whether warm or cool tones are used (Figure 18-25). Many color product lines have a specific gray coverage series that provides greater saturation of the color.

Occasionally, gray hair is so resistant that pre-softening is necessary for better color penetration. It is always a good idea to mix a small amount of the haircolor product and perform a strand test. A strand test indicates:

- if the hair is resistant to the product
- how long it takes for the product to be absorbed by the hair
- what the color results will be

RECONDITIONING DAMAGED HAIR

Hair that is damaged due to careless chemical applications, excessive heat, or misused styling products must be reconditioned before it can be tinted or lightened successfully. Sometimes hair is naturally brittle, thin, and lifeless. Both neglect and the client's physical condition may contribute to these conditions. Hair is considered damaged when it exhibits one or more of the following characteristics:

- overporous
- brittle and dry
- breaks easily
- little to no elasticity
- rough and harsh to the touch
- spongy and mats easily when wet
- resists color or absorbs too much color during a tinting process

Any of these conditions may create undesirable results during a tinting or lightening service. Therefore, damaged hair should receive reconditioning treatments before and after haircolor applications.

figure 18-25

Skin tone, eye color, and personal preference will determine if warm or cold tones are used.



Reconditioning Treatment

To restore damaged hair to a healthier condition, hair conditioners containing lanolin or protein substances should be used. The conditioning product is applied to the hair according to the manufacturer's directions. If heat is applied, use a heating cap, a steamer, or a heating lamp. Be guided by your instructor as to the frequency and length of time for each treatment.

TINT BACK TO NATURAL COLOR

Clients who have been tinting or lightening their hair may want to return to their natural shade, which is known as a *tint back*. Each tint back must be handled as an individual situation. The determining factors in the selection of the tint shade are the:

- present condition of the hair
- present color of the hair
- final result desired
- original color

To determine the client's natural hair color, check the natural shade of the hair next to the scalp. Next, select an appropriate shade of filler to correspond with the tint to be used; otherwise, it will be difficult to obtain a uniform color from the scalp to hair ends, due to uneven porosity levels. Perform strand tests as needed to determine the expected final outcome.

COLORING MUSTACHES AND BEARDS

- *Aniline derivative tints* should *never* be used for coloring mustaches. Doing so may cause serious irritation or damage to the lips or the delicate membranes of the nostrils.
- *Metallic or progressive dyes* have been known to cause severe allergic reactions when applied to facial hair or around freshly cut hairlines.
- *Hair color crayons* are waxy sticks that are available in several colors: blond, medium, dark brown, black, and auburn. They are applied by rubbing it directly on the facial hair.
- *Pomades* are formulated specifically for coloring mustaches and beards. These products are available in a variety of shades, including black, brown, blond, chestnut, and white (neutral). The pomade is applied with a small brush and is stroked from the nostrils downward. Liquid pomades are also available and may be preferred for use on beards. Some pomades contain heavy waxing ingredients that can be used to style mustaches with rolled or twisted ends for dramatic looks.
- *Liquid eyebrow and eyelash tints* are available in brown and black for coloring facial hair.

P 18-8 Coloring Mustaches and Beards *pages 699-701*

COATING DYES

Many clients buy and use over-the-counter haircoloring products at home. Some of these products are actually progressive dyes and must be removed prior to any other chemical service. Hair treated with a compound, metallic, or other coating dye looks dry and dull, and may feel harsh and brittle to the touch. These colors usually fade to unnatural tones, such as:

- silver dyes have a green or gray cast
- lead dyes leave a purple color
- those containing copper turn red

If you are unsure as to whether the client has used a progressive dye, a test for metallic salts and dyes should be performed on the hair.

Test for Metallic Salts and Coating Dyes

In a glass container, mix 1 ounce of 20-volume (6 percent) peroxide and 20 drops of 28 percent ammonia water. Cut a few strands of the client's hair, bind it with tape, and immerse it in the solution for 30 minutes. Remove, towel-dry, and observe the strand. Refer to the following for analysis of the hair:

- Hair dyed with lead will lighten immediately.
- Hair treated with silver will show no reaction at all, which indicates that other chemicals will not be successful because they will not be able to penetrate the coating.
- Hair treated with copper will start to boil and will pull apart easily. This hair would be severely damaged or destroyed if other chemicals such as those found in permanent colors or perm solutions were applied to it.
- Hair treated with a coating dye either will not change color or will lighten in spots. Hair in this condition will not receive chemical services easily and the length of time necessary for penetration may result in further damage to the hair.

Removing Coatings from the Hair Mini-Procedure

The removal of metallic dyes from the hair shaft may not always be effective the first time. Performing a strand test after the treatment will indicate whether the metallic deposits have been removed. If not, the entire application must be repeated until the hair is sufficiently free of metal salts to perform other chemical services.

Supplies

- ☐ 70 percent alcohol
- ☐ Concentrated shampoo for oily hair
- ☐ Mineral, castor, vegetable, or commercially prepared color-removing oil



REMINDER

Keep up to date! Manufacturers are constantly improving and developing new haircoloring products. Be sure to attend seminars and trade shows as often as possible to stay current in your profession.

Procedure

1. Apply 70 percent alcohol to dry hair.
2. Allow alcohol to stand for 5 minutes.
3. Apply the oil to the hair thoroughly.
4. Cover the hair completely with a plastic cap.
5. Place under a hot dryer for 30 minutes.
6. To remove the oil, saturate with concentrated shampoo.
7. Work the shampoo into the oil for 3 minutes, and then rinse with warm water.
8. Repeat the shampoo steps until the oil is removed completely.

Discuss Haircoloring and Lightening Safety Precautions

After reading this section, you will be able to:

- LO 7** List haircoloring and lightening safety precautions.

HAIRCOLORING SAFETY PRECAUTIONS

- Perform a 24- to 48-hour patch test before the application of a haircolor product containing aniline derivative.
- Examine the scalp before applying a tint.
- Do not apply tint if abrasions are present on the scalp.
- Use only clean swabs, brushes, applicator bottles, combs, and linens.
- Always wash your hands before and after each client.
- Do not brush the hair prior to a tint.
- Do not apply a tint without reading the manufacturer's directions.
- Perform a strand test for color and processing results.
- Choose a shade of tint that harmonizes with the client's complexion.
- Use an applicator bottle or bowl (plastic or glass) for mixing tint.
- Do not mix tint before ready for use; discard leftover tint.
- If required, use the correct shade of color filler.
- Make frequent strand tests until the desired shade is reached.
- Suggest a reconditioning treatment for tinted hair.
- Do not apply tint if metallic or compound dye is present.
- Do not apply tint if a patch test is positive.

- Perform a strand test for the correct color shade before applying tint.
- Do not use an alkaline or harsh shampoo for tint removal.
- Do not use water that is too hot for removing tint. Use lukewarm or tepid water.
- Protect the client's clothing by proper draping.
- Do not permit tint to come in contact with the client's eyes.
- Do not overlap during a tint retouch.
- Cap all bottles of developer and tint to avoid loss of strength.
- Fill out a tint record card.
- Do not apply hydrogen peroxide or any material containing hydrogen peroxide directly over dyes known or believed to contain a metallic salt. Breakage or complete disintegration of the hair may result.
- Wear protective gloves.

HAIR LIGHTENING SAFETY PRECAUTIONS

- Analyze the condition of the hair and suggest reconditioning treatments, if required.
- When working with a cream or paste lightener, it must be the thickness of whipped cream to avoid dripping or running and overlapping.
- Do not use off-the-scalp lightener for virgin or retouch lightener applications.
- Apply lightener to resistant areas first. Use 1/8-inch sections when applying lightener to ensure complete coverage.
- Check strands frequently until the desired lightness is reached.
- Immediately after completing the lightener application, check the client's skin (face, neck, hairline, etc.) and remove any lightener to avoid skin irritation.
- Check the towel around the client's neck. Lightener on the towel that is allowed to come in contact with the skin will cause irritation.
- Lightened hair is fragile and requires special care. Use only a very mild shampoo and only cool water for rinsing.
- If a preliminary shampoo is necessary, comb and shampoo the hair carefully. Avoid irritating the scalp during the shampoo or when combing the hair.
- Work as rapidly as possible when applying the lightener to produce a uniform shade without streaking.
- Never allow lightener to stand; use it immediately.
- Cap all bottles of developer and lightener to avoid loss of strength.
- Keep a completed record card of all lightening treatments.

PATCH TEST

MATERIALS, IMPLEMENTS, AND EQUIPMENT

- ☐ Cotton swab
- ☐ Developer
- ☐ Glass or plastic mixing bowl
- ☐ Haircolor product
- ☐ Mild soap
- ☐ Towels

PREPARATION

- 1 Wash your hands.
- 2 Conduct client consultation and select a haircolor product.
- 3 Perform hair and scalp analysis.
- 4 Select and arrange required materials.
- 5 Drape client for a chemical service.

PROCEDURE



- 1 Select a test area; behind the ear or on the inside of the elbow are good choices.
- 2 Using a mild soap, clean and dry an area about the size of a quarter.



- 3 Mix a small amount of the same product you plan on using for the service according to the manufacturer's directions.



- ④ Apply a small amount of the haircolor mixture to the test area with a sterile cotton swab.
- ⑤ Leave the mixture undisturbed for 24 to 48 hours.
- ⑥ Examine the test area. If there are no signs of redness or irritation, the test result is negative, and you can proceed with the color service.



- ⑦ Record the results on the haircolor service record card.

CLEAN-UP AND DISINFECTION

- ☐ Wash and disinfect tools and implements.
- ☐ Clean and disinfect work area.
- ☐ Dispose of single-use items. Place all used linens, towels, and capes in the laundry.
- ☐ Wash your hands.

STRAND TEST

MATERIALS, IMPLEMENTS, AND EQUIPMENT

- ☐ Bowl and brush
- ☐ Chemical cape
- ☐ Color brushes
- ☐ Developer
- ☐ Glass or plastic mixing bowl
- ☐ Protective gloves
- ☐ Selected haircolor
- ☐ Service record card
- ☐ Shampoo
- ☐ Sheet of foil or plastic wrap
- ☐ Spray bottle containing water
- ☐ Timer
- ☐ Towels

PREPARATION

- 1 Wash your hands.
- 2 Conduct client consultation and select a haircoloring product.
- 3 Perform hair and scalp analysis.
- 4 Select and arrange required materials.
- 5 Drape client for a chemical service.

PROCEDURE



- 1 Apply gloves. Part off a ½-inch square section of hair in the interior nape area, so it is not visible from the hairline. Using plastic clips, fasten other hair out of the way.



- 2 Place the hair strand over the foil or plastic wrap and apply the color mixture you plan on using for the service.